

Higher Categories I

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Introduction

About This Document

TODO

A Fairytale

At many places in homotopy theory interesting maps are uniquely determined up to homotopy but not on the nose. Let me give 4 examples:

- 1) The inverse to a homotopy equivalence
- 2) The classifying map $B \rightarrow BO(d)$ of a d -dimensional vector bundle
- 3) The lift in a HELP-diagram with one of the vertical maps a homotopy equivalence

$$\begin{array}{ccc}
 A & \xrightarrow{\quad} & E \\
 \text{cofibration} \downarrow & \nearrow & \downarrow \text{fibration} \\
 X & \xrightarrow{\quad} & B
 \end{array}$$

- 4) The n -fold multiplication

$$(\Omega X)^n \rightarrow \Omega X$$

Similar phenomena occur in algebra:

- 5) The map $F \rightarrow F'$ between free resolutions of $M \rightarrow N$.

Now, faced with such a situation it is often difficult to perform categorically flavoured operations in a meaningful way:

A simple example is the formation of fibres: f^{-1} is the pullback

$$\begin{array}{ccc}
 f^{-1}(y) & \xrightarrow{\quad} & X \\
 \downarrow & & \downarrow f \\
 y & \xrightarrow{\quad i \quad} & Y
 \end{array}$$

but homotopic f and f' (or homotopic i, i') need not give homotopy equivalent pullbacks. To assign invariant meaning to them homotopical versions of such constructions were soon invented, in the case at hand, the *homotopy fibre*.

But this still not quite good enough:

- 1) While homotopy fibres of homotopic maps f and f' are homotopy equivalent, the (homotopy class of) homotopy equivalence depends on a choice of (homotopy class of) homotopy between f and f' , so the homotopy fibre is not functorial on the arrow category of the homotopy category.
- 2) The homotopy fibre is not a direct categorical construction anymore neither in the original category nor the homotopy category; it represents neither the functor of commuting diagrams nor of homotopy commuting diagrams. Rather a map

$$Z \rightarrow \text{hofib}_x(f)$$

represents a *homotopy coherent* diagram

$$\begin{array}{ccc}
 Z & \xrightarrow{\quad} & X \\
 \downarrow & \searrow & \downarrow f \\
 * & \xrightarrow{x} & Y
 \end{array}$$

(double arrows indicating homotopies)

The first problem is addressed by the observation that the maps in question are all determined up to contractible choice not just up to homotopy! Let me warn you that it is not true in examples 1), 2), 4) and 5) that the component of a given map in the mapping space is contractible, just that there is a contractible "space", in which the maps in question naturally live: for example in 2) the space of bundle maps

$$V \rightarrow EO(d) \times_{O(d)} \mathbb{R}^d$$

is contractible, in 4) the multiplications are given partitions of the interval into 4 parts i.e. the space

$$\{(t_1, t_2, t_3) \in I^3 \mid 0 < t_1 < t_2 < t_3 < 1\}$$

parametrizes the multiplications and is contractible. For a homotopy equivalence $g : Y \rightarrow X$ it is a space of pairs

$$\{(f, H) \in C(X, Y) \times C(X \times I, X) \mid H_0 = \text{id}_X, H_1 = g \circ f\} .$$

In particular, for any two choices of map as in 1)-5) not only does there exist a homotopy between them, but that homotopy is itself unique up to homotopy (so for example for a vector bundle $V \rightarrow B$ the homotopy fibre

$$\text{hofib}_\delta(B \rightarrow BSO(d))$$

is well-defined up to unique isomorphism in the homotopy category (once a point $\delta \in BSO(d)$ is fixed once and for all). But moreover any two homotopies witnessing this uniqueness are again unique up to homotopy, so further constructions of similar flavour are again possible without destroying functoriality. To exploit these observations a systematic category theory allowing for homotopy and contractibility instead of equality and uniqueness is required. The easiest way to achieve this is by considering topologically enriched categories. To

accommodate non-topological examples it is in fact better to work combinatorially and consider Kan-enriched categories. While the entirety of higher category theory can be developed for these objects, they carry a major defect: composition of morphisms is still strictly associative. While this is the case for most categories that serve as starting points ($\mathbf{Top}$, $\mathbf{sSet}$, $\mathbf{Ch}_R$, ...), it is not preserved by many categorical constructions, since (by design) these are only determined up to contractible choice. A related problem is that a functor $F: \mathcal{C} \rightarrow \mathcal{D}$ of simplicial categories ought to be considered invertible if

- 1) it is essentially homotopically surjective, meaning for all $D \in \mathcal{D}$ there is a $C \in \mathcal{C}$ and a homotopy equivalence $F(C) \simeq D$ and
- 2) F induces a homotopy equivalence

$$\mathrm{Hom}(A, B) \rightarrow \mathrm{Hom}(FA, FB)$$

for all objects $A, B \in \mathcal{C}$.

Such functors are called *Dwyer-Kan equivalences*.

Although this notion restricts to usual equivalences of categories for discretely enriched $\mathcal{C}$ and $\mathcal{D}$ it is quite different from that of an enriched equivalence, which would require isomorphisms

$$\mathrm{Hom}(A, B) \rightarrow \mathrm{Hom}(FA, FB).$$

Here are two examples. Consider the full subcategory $\mathcal{C}$ of the simplicially enriched category $\mathbf{Top}$ spanned by contractible spaces. Then the functor $p: \mathcal{C} \rightarrow *$ to the one-point category (with trivial enrichment) is a Dwyer-Kan equivalence, since the space of continuous functions between contractible spaces is itself contractible. However, for any functor $s: * \rightarrow \mathcal{C}$ a natural transformation $\tau: s \circ p \Rightarrow \mathrm{id}_{\mathcal{C}}$ would require all diagrams of the form

$$\begin{array}{ccc} s(*) & \xrightarrow{\tau_X} & X \\ (s \circ p)(f) \downarrow & & \downarrow f \\ s(*) & \xrightarrow{\tau_Y} & Y \end{array}$$

to commute, which is clearly absurd since $s(*) \neq \emptyset$. Note, however, that a (unique) transformation $\mathrm{id}_{\mathcal{C}} \Rightarrow s \circ p$ exists just fine, showing that natural transformations, which are pointwise equivalences need not admit inverses, another desirable criterion that Kan-enriched categories fit. As a second example consider the category $\mathcal{N}[2]$ with three objects and six morphisms

We can give it the trivial enrichment. Consider also the simplicially enriched category $\mathcal{C}$, whose only difference is that we replace

$$\mathrm{Hom}_{\mathcal{N}[2]}(0, 2) = \mathrm{const}\{f, g\} \in \mathbf{sSet}$$

by

$$\mathrm{Hom}_{\mathcal{C}}(0, 2) = \mathrm{Sing} \, \mathbf{I}$$

with $g \circ f = (\Delta^0 \xrightarrow{0} \mathbf{I}) \in \mathrm{Sing}_0 \, \mathbf{I}$. Evidently, there are unique enriched functors

$$\mathcal{C} \rightarrow \mathcal{N}[2] \quad \text{and} \quad \mathcal{N}[2] \rightarrow \mathcal{C}$$

which are identity on objects and both are Dwyer-Kan equivalences. However, only one make the diagram

$$\begin{array}{c}
 \text{id} \circlearrowleft \bullet^0 \longrightarrow \bullet^1 \circlearrowright \text{id} \quad \equiv \quad \mathcal{N}[1] \quad \begin{array}{c} \xrightarrow{\begin{smallmatrix} 0 & 1 \\ 1 & 2 \end{smallmatrix}} \mathcal{N}[2] \\ \xrightarrow{\begin{smallmatrix} 0 & 1 \\ 1 & 2 \end{smallmatrix}} \mathcal{C} \end{array} \\
 \Delta^0 \xrightarrow{1} I
 \end{array}$$

commute. The worst offender, however, are Dwyer-Kan equivalences which do not admit inverses at all: for example, there is no reason to expect that the Dwyer-Kan equivalence

$$\left\{ \begin{array}{c} \text{finite cell complexes,} \\ \text{cellular}^2 \text{maps} \end{array} \right\} \subseteq \left\{ \begin{array}{c} \text{spaces homotopy equivalent to a finite cell complex,} \\ \text{continuous maps} \end{array} \right\}$$

admits any functor the other way that induces the inverse on homotopy categories. To remedy these deficits one considers a notion of higher category theory that is not equipped with a strict composition law. There are at least two good candidates: Joyal's *quasi-categories* and Rezk's *complete Segal spaces*. Given the recent development of the former at the hands of Lurie we will give preference to it and say little about the later notion. Quasi-categories were originally defined by Boardman and Vogt (who taught me topology in Osnabrück) during the development of E_∞ -spaces (the natural notion of commutative monoid that arises in higher categories): they attempted to define the morphisms between such objects as maps of the underlying spaces together with an E_∞ -structure on its cofibre extending the given ones. As cofibres do not commute with composition they did not quite obtain a category and instead formalized the resulting object (under the name "weak Kan complex"). This apparent defect was soon remedied by May's reformulation of their theory via his introduction of operads and thus stayed undeveloped for many years until Joyal demonstrated that much of ordinary category theory generalises to such weak Kan complexes (whence he renamed them). A quasi-category is (roughly speaking) a simplicial set $\mathcal{C}$, which admits compositions up to contractible choice; here one thinks of $\mathcal{C}_0$ as the objects and $\mathcal{C}_1$ as the morphisms of our higher category. The space of composites of two 1-simplices $f, g \in \mathcal{C}_1$ is then the pullback

$$\begin{array}{ccc}
 \text{Comp}(f, g) & \longrightarrow & \text{Hom}(\Delta^2, \mathcal{C}) \\
 \downarrow & & \downarrow \\
 * & \xrightarrow{(f, g)} & \text{Hom}(\Lambda_1^2, \mathcal{C})
 \end{array}$$

Warning: the actual definition requires the right hand vertical map to also be a Kan fibration!

We will develop the very basics of this notion. The main results we will hopefully show, say precisely that quasi-categories do not share the defects of Kan-enriched categories, i.e. that

⁴Suitably interpreted as a simplicial set the statement fails for the naive subspace of cellular maps inside all maps!

- 1) functors are equivalences if they are fully faithful and essentially surjective (both in the homotopical sense) and
- 2) natural transformations are invertible if they consist of equivalences.

Furthermore, we will show that the classical homotopy theory of cell complexes is precisely the category theory of higher groupoids, a fact known as Grothendieck's *homotopy hypothesis*.

By way of compensation it is, however, quite a bit more difficult to produce interesting quasi-categories directly. We will therefore develop the *homotopy coherent nerve*, which produces a quasi-category from a Kan-enriched ordinary category. Both these last points require a bit a classical simplicial homotopy theory (which is the theory of Kan complexes) so we will treat that first.

Finally, let me just mention that the theories just discussed are only the start: they are all on models for $(\infty, 1)$ -category theory, meaning that all morphisms of degree ≥ 2 (starting with homotopies) are automatically invertible. For example, the (large) $(\infty, 1)$ -category Cat_∞ of (small) $(\infty, 1)$ -categories really wants to be an $(\infty, 2)$ -category (since not all natural transformations are equivalences).

We will have little to say on this. Other topics that will not make an appearance here:

- straightening / unstraightening
- Yoneda's Lemma
- operads / symmetric monoidal categories
- sheaves / hypersheaves
- spectra and stabilization
- systematic study of limits and colimits

Maybe if there is a part 2...

Chapter I.

Categories

I.1. Definition

A *category* $\mathcal{C}$ consists of two classes of O and M whose elements are called *objects* and *morphisms*, respectively, and four maps

$$i : O \rightarrow M, \quad s : M \rightarrow O, \quad t : M \rightarrow O$$

and

$$c : M \times_t M \rightarrow M$$

called *identity*, *source*, *target* and *composition* such that

- | | |
|--------------------------------------|--|
| 1) $s \circ i = \text{id}_O$ | 3) $c \circ (c \times_O \text{id}_M) = c \circ (\text{id}_M \times_{OC})$ |
| $t \circ i = \text{id}_O$ | |
| 2) $s \circ c = s \circ \text{pr}_2$ | 4) $c \circ ((i \circ t) \times \text{id}_M) \circ \Delta_M = \text{id}_M$ |
| $t \circ c = t \circ \text{pr}_1$ | $c \circ (\text{id}_M \times (i \circ s)) \circ \Delta_M = \text{id}_M$ |

For $x, y \in O$ we set

$$\text{Hom}_{\mathcal{C}}(x, y) = \{\varphi \in M \mid s(\varphi) = x, t(\varphi) = y\}$$

$\mathcal{C}$ is called *small* if both O and M are sets and *locally small* if $\text{Hom}_{\mathcal{C}}(x, y)$ is a set for every $x, y \in O$. More generally an object $x \in O$ is called *left* or *right locally small* if $\text{Hom}_{\mathcal{C}}(x, y)$ or $\text{Hom}_{\mathcal{C}}(y, x)$ are (in bijection with) sets for all $y \in O$.

For a category $\mathcal{C} = (O, M, \dots)$ one writes

$$x \in \mathcal{C}, \varphi : x \rightarrow y, \text{id}_x, \varphi \circ \psi$$

instead of

$$x \in O, \varphi \in \text{Hom}_{\mathcal{C}}(x, y), i(x), c(\varphi, \psi).$$

I.2. Remark

By 1) smallness is implied by M being a set, and for locally small categories also by the smallness of O .

I.3. Definition

A morphism $\varphi : x \rightarrow y$ is called a *split monomorphism*, or *split epimorphism* if it admits a *retraction* $\psi : y \rightarrow x$, that is $\psi \circ \varphi = \text{id}_x$, or a *section*, that is $\varphi \circ \psi = \text{id}_y$, respectively. It is called an *isomorphism* if it admits an *inverse*, that is $\psi : y \rightarrow x$ that is both a retraction and a section simultaneously.

I.4. Definition

A category $\mathcal{C}$ is called *essentially small* with *skeleton* S , a subset of O , if it is locally small and for each $x \in O$ there exists a $y \in S$ and an isomorphism $\varphi : x \rightarrow y$.

I.5. Definition

A functor $F : \mathcal{C} \rightarrow \mathcal{C}'$ is a pair

$$F : O \rightarrow O' \quad , \quad F_* : M \rightarrow M'$$

that commutes with all structure maps, i.e. such that

$$\begin{aligned} 1) \quad F_* \circ i &= i \circ F & 2) \quad s' \circ F_* &= F \circ s \\ & & t' \circ F_* &= F \circ t \end{aligned}$$

$$3) \quad F_* \circ c = c \circ (F_* \times_O F_*)$$

We denote the class of functors $\mathcal{C} \rightarrow \mathcal{C}'$ by $\text{Fun}(\mathcal{C}, \mathcal{C}')$. F is called *full* or *faithful* if the map

$$(F_*, s, t) : M \rightarrow M' \times_{O' \times O'} O \times O^1$$

is surjective or injective, respectively. An *embedding* is a functor that is fully faithful.

I.6. Examples

i) Set, Top, Ring, R-Mod, Grp, Ab, Ord, Cat, ...

ii) For $\mathcal{M}$ a monoid there is the category $\mathbb{B}\mathcal{M}$ with

$$O = \{*\} \text{ and } M = \mathcal{M}$$

$i(*) = e$, and composition the multiplication in $\mathcal{M}$

iii) For every poset P there is the category NP with

$$O = P \text{ and } M = \{(i, j) \in P^2 \mid i \leq j\}$$

$i = \Delta_P, s = \text{pr}_1, t = \text{pr}_2$ and

$$c((p, q), (r, p)) = (r, q)$$

¹ $\{(x, y, \varphi) \in O^2 \times M' \mid \varphi : F(x) \rightarrow F(y)\}$

Chapter I. Categories

For $[n] = \{0, 1, 2, \dots, n\}$ we find $N[n]$ pictorially

$$\bullet \xrightarrow{(0,1)} \bullet \xrightarrow{(1,2)} \bullet \longrightarrow \dots \quad + \begin{array}{l} \text{composites } (i, j) \quad i < j + 1 \\ \text{identities } (i, i) \end{array}$$

We obtain an embedding $N : \text{Ord} \rightarrow \text{Cat}$.

iv) The various $\text{Hom}_{\mathcal{C}}(x, y)$ assemble into a functor $\text{Hom}_{\mathcal{C}} : \mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \text{Set}$.

I.7. Definition

A *natural transformation* $\tau : F \Rightarrow G$ of functors $\mathcal{C} \rightarrow \mathcal{C}'$ is a map

$$\tau : O \rightarrow M', \text{ s.t.}$$

$$\begin{array}{ll} 1) \quad s' \circ \tau = F & 2) \quad c' \circ (G \times (\tau \circ s)) \circ \Delta_M \\ & = c' \circ ((\tau \circ t) \times F) \circ \Delta_M \\ & t' \circ \tau = G \end{array}$$

From now on we will write $\text{obj}(\mathcal{C}) := O$ and $\text{mor}(\mathcal{C}) := M$.

I.8. Definition

Denote by $d_i : [n] \rightarrow [n+1]$ ($0 \leq i \leq n+1$) and $s_i : [n] \rightarrow [n-1]$ ($0 \leq i \leq n-1$) the unique order preserving injection and surjection, respectively, that misses i or hits i twice.

I.9. Definition

For categories $\mathcal{C}, \mathcal{C}'$ define the *category of functors* $\mathcal{C} \rightarrow \mathcal{C}'$ to have objects and morphisms

$$\text{Fun}(\mathcal{C} \times N[0], \mathcal{C}') \text{ and } \text{Fun}(\mathcal{C} \times N[1], \mathcal{C}')$$

respectively; structure maps are

$$i = - \circ (\text{id}_{\mathcal{C}} \times N_* s_0), \quad s = - \circ (\text{id}_{\mathcal{C}} \times N_* d_1), \quad t = - \circ (\text{id}_{\mathcal{C}} \times N_* d_0)$$

and composition is given by

$$\begin{array}{ccc} \text{Fun}(\mathcal{C} \times N[2], \mathcal{C}') & \xrightarrow{- \circ N_* d_1} & \text{Fun}(\mathcal{C} \times N[1], \mathcal{C}') \\ \downarrow \cong & & \\ \text{Fun}(\mathcal{C} \times N[1], \mathcal{C}') \times_{\text{Fun}(\mathcal{C} \times N[0], \mathcal{C}')} & & \text{Fun}(\mathcal{C} \times N[1], \mathcal{C}') \end{array}$$

Warning. If $\mathcal{C}$ is small and $\mathcal{C}'$ locally small then $\text{Fun}(\mathcal{C}, \mathcal{C}')$ is locally small, but this can fail when $\mathcal{C}$ is only locally small!

I.10. Definition

A natural transformation that is an isomorphism in $\text{Fun}(\mathcal{C}, \mathcal{C}')$ is called a *natural isomorphism*.

I.11. Definition

A functor $F : \mathcal{C} \rightarrow \mathcal{C}'$ is called an *equivalence* if there exists a functor $G : \mathcal{C}' \rightarrow \mathcal{C}$ and natural isomorphisms

$$F \circ G \Rightarrow \text{id}_{\mathcal{C}'} \quad \text{and} \quad G \circ F \Rightarrow \text{id}_{\mathcal{C}}.$$

I.12. Definition

The *essential image* $F(\mathcal{C})$ of a functor $F : \mathcal{C} \rightarrow \mathcal{C}'$ is the full² subcategory generated by all objects isomorphic to one in the image of $F : \mathcal{O} \rightarrow \mathcal{O}'$.

I.13. Warning

The actual image

$$(F(\mathcal{O}), F_*(M)) \subseteq (\mathcal{O}', M')$$

is not generally closed under composition in $\mathcal{C}'$, so does not form a subcategory. The notion of equivalence of categories is *not* that of isomorphism in Cat , which requires $F \circ G = \text{id}_{\mathcal{C}'}$ and $G \circ F = \text{id}_{\mathcal{C}}$!

I.14. Proposition

- i) A functor $F : \mathcal{C} \rightarrow \mathcal{C}'$ is an equivalence if and only if it is an embedding and essentially surjective, i.e. $F(\mathcal{C}) = \mathcal{C}'$.
- ii) A natural transformation is a natural isomorphism if and only if it takes values in isomorphisms.

Proof. MacLane ...

□

I.15. Corollary

Every embedding of categories is an equivalence onto its essential image³. For example, the inclusion of every full subcategory $\mathcal{C} \subseteq \mathcal{C}'$ containing an object from every isomorphism class of $\mathcal{C}'$ is an equivalence.

²i.e. with morphisms the entire Hom-sets.

³It need not be an isomorphism onto its image!

I.16. Definition

A category is called a *groupoid* if every morphism in it is an isomorphism. A typical example is the inclusion of the *groupoid core*

$$\text{Core}(\mathcal{C}) \subseteq \mathcal{C}$$

having the same objects as $\mathcal{C}$ but retaining only the isomorphisms of $\mathcal{C}$ as morphisms. Also the functor $\mathbb{B}$ factors as $\text{Grp} \xrightarrow{\mathbb{B}} \text{Grpd} \subseteq \text{Cat}$.

Every left/right locally small object $x \in \mathcal{C}$ determines the functors

$$\begin{array}{l|l} \text{Hom}_{\mathcal{C}}(x, -) = Y_x : \mathcal{C} \rightarrow \text{Set} & \text{Hom}_{\mathcal{C}}(-, x) = Y^x : \mathcal{C}^{\text{op}} \rightarrow \text{Set} \\ y \mapsto \text{Hom}_{\mathcal{C}}(x, y) & y \mapsto \text{Hom}_{\mathcal{C}}(y, x) \\ f \mapsto f \circ - & f \mapsto - \circ f \end{array}$$

known as the functors *(co)represented* by x .

The arguably most important statement of category theory is

I.17. Proposition (Yoneda's Lemma)

For all functors $F : \mathcal{C} \rightarrow \text{Set}$ and $G : \mathcal{C}^{\text{op}} \rightarrow \text{Set}$ and $x \in \mathcal{C}$ left or right locally small, the maps $\text{Nat}(Y_x, F) \rightarrow F(x)$ and $\text{Nat}(Y^x, G) \rightarrow G(x)$ given by

$$\tau \mapsto \tau_x(\text{id}_x)$$

are bijections, respectively.

Proof. MacLane ...

□

For locally small $\mathcal{C}$, the functors Y^x give rise to a single functor

$$Y^{\mathcal{C}} : \mathcal{C} \rightarrow \text{Fun}(\mathcal{C}^{\text{op}}, \text{Set}) =: \mathcal{P}(\mathcal{C})^4$$

for example, as the image of $\text{Hom}_{\mathcal{C}} \in \text{Fun}(\mathcal{C}^{\text{op}} \times \mathcal{C}, \text{Set})$ under

$$\text{Fun}(\mathcal{C}^{\text{op}} \times \mathcal{C}, \text{Set}) \cong \text{Fun}(\mathcal{C}, \text{Fun}(\mathcal{C}^{\text{op}}, \text{Set}))$$

I.18. Corollary/Definition

Let $\mathcal{C}$ be locally small. The *Yoneda embedding*

$$Y^{\mathcal{C}} : \mathcal{C} \rightarrow \text{Fun}(\mathcal{C}^{\text{op}}, \text{Set})$$

is really an embedding. Its essential image is called the category of *representables*⁵ of $\mathcal{C}$.

⁴not usually locally small

⁵they are left locally small

Thus one can specify a functor $\mathcal{C} \rightarrow \mathcal{D}$ (with $\mathcal{D}$ locally small) up to unique isomorphism by giving a functor

$$\mathcal{C} \rightarrow \text{Fun}(\mathcal{D}^{\text{op}}, \text{Set})$$

and checking that its image is contained in the representables. Specialized to $\mathcal{C} = *$ this is a method of definition by *universal property*!

An example is:

I.19. Definition

An object $x \in \mathcal{C}^6$ is *left* or *right adjoint* to an object $y \in \mathcal{D}^6$ under a functor $F : \mathcal{C} \rightarrow \mathcal{D}$, if, respectively, there is a natural isomorphism

$$\tau^7 : Y_x \Longrightarrow Y_y \circ F \quad \text{or} \quad \tau^8 : Y^x \Longrightarrow Y^y \circ F^{\text{op}},$$

which is called a *witness* of the adjunction.

I.20. Examples

- i) The integers are left adjoint to the empty set under the forgetful functor $\text{Ring} \rightarrow \text{Set}$.
- ii) An R -module is free if and only if it is left adjoint to some set under the forgetful functor $R\text{-Mod} \rightarrow \text{Set}$ and a witness is precisely a choice of basis indexed by that set.
- iii) The groupoid core of $\mathcal{C}$ is right adjoint to $\mathcal{C}$ under the inclusion $\text{Grpd} \subseteq \text{Cat}$.
- iv) The set

$$\pi_0(\mathcal{C}) = \text{obj}(\mathcal{C}) / \sim,$$

where $\sim$ is the equivalence relation generated by setting

$$x \sim y \quad \text{if} \quad \text{Hom}_{\mathcal{C}}(x, y) \neq \emptyset,$$

is left adjoint to $\mathcal{C}$ under $\mathbb{D} : \text{Set} \rightarrow \text{Cat}$

If $\mathcal{D}_F \subseteq \mathcal{D}$ and $\mathcal{D}^F \subseteq \mathcal{D}$ denote the full subcategories spanned by the objects admitting a left or right adjoint, respectively, then the functors

$$\mathcal{D}_F^{\text{op}} \subseteq \mathcal{D}^{\text{op}} \xrightarrow{Y_{\mathcal{D}}} \text{Fun}(\mathcal{D}, \text{Set}) \xrightarrow{- \circ F} \text{Fun}(\mathcal{C}, \text{Set})$$

$$\mathcal{D}^F \subseteq \mathcal{D} \xrightarrow{Y^{\mathcal{D}}} \text{Fun}(\mathcal{D}^{\text{op}}, \text{Set}) \xrightarrow{- \circ F^{\text{op}}} \text{Fun}(\mathcal{C}^{\text{op}}, \text{Set})$$

take values in the (co)representables. Thus they determine functors

$$L_F : \mathcal{D}_F \rightarrow \mathcal{C} \quad \text{and} \quad R_F : \mathcal{D}^F \rightarrow \mathcal{C}$$

together with natural isomorphisms

$$\text{Hom}_{\mathcal{C}} \circ (L_F^{\text{op}} \times \text{id}_{\mathcal{C}}) \Longrightarrow \text{Hom}_{\mathcal{D}} \circ (\text{id}_{\mathcal{D}_F^{\text{op}}} \times F)$$

⁶locally small, but see I.23

⁷i.e. $\text{Hom}_{\mathcal{C}}(x, -) \cong \text{Hom}_{\mathcal{D}}(y, F(-))$

⁸i.e. $\text{Hom}_{\mathcal{C}}(-, x) \cong \text{Hom}_{\mathcal{D}}(F(-), y)$

and

$$\mathrm{Hom}_{\mathcal{C}} \circ (\mathrm{id}_{\mathcal{C}^{\mathrm{op}}} \times R_F) \implies \mathrm{Hom}_{\mathcal{D}} \circ (F^{\mathrm{op}} \times \mathrm{id}_{\mathcal{D}^F})$$

of functors

$$\mathcal{D}_F^{\mathrm{op}} \times \mathcal{C} \rightarrow \mathrm{Set} \quad \text{and} \quad \mathcal{C}^{\mathrm{op}} \times \mathcal{D}^F \rightarrow \mathrm{Set},$$

respectively. In case F happens to take values in $\mathcal{D}_F$ or $\mathcal{D}^F$ the functors L_F or R_F also determine F as R_{L_F} or L_{R_F} , respectively. We set:

I.21. Definition

An *adjunction* between functors $L : \mathcal{C} \rightarrow \mathcal{D}$ and $R : \mathcal{D} \rightarrow \mathcal{C}$ is a natural isomorphism

$$\tau : \mathrm{Hom}_{\mathcal{D}} \circ (L^{\mathrm{op}} \times \mathrm{id}_{\mathcal{D}}) \implies \mathrm{Hom}_{\mathcal{C}} \circ (\mathrm{id}_{\mathcal{C}^{\mathrm{op}}} \times R).$$

One often says that τ exhibits L as left adjoint to R or R as right adjoint to L in this situation.

I.22. Proposition/Definition

Any adjunction is both of the form

$$\mathrm{Hom}_{\mathcal{D}} \circ (L^{\mathrm{op}} \times \mathrm{id}_{\mathcal{D}}) \xRightarrow{R_*^9} \mathrm{Hom}_{\mathcal{C}} \circ (R^{\mathrm{op}} L^{\mathrm{op}} \times R) \xRightarrow{u^*} \mathrm{Hom}_{\mathcal{C}} \circ (\mathrm{id}_{\mathcal{C}^{\mathrm{op}}} \times R)$$

and inverse to

$$\mathrm{Hom}_{\mathcal{C}} \circ (\mathrm{id}_{\mathcal{C}^{\mathrm{op}}} \times R) \xRightarrow{L_*^{10}} \mathrm{Hom}_{\mathcal{D}} \circ (L^{\mathrm{op}} \times LR) \xRightarrow{c_*} \mathrm{Hom}_{\mathcal{D}} \circ (L^{\mathrm{op}} \times \mathrm{id}_{\mathcal{D}})$$

for unique natural transformations

$$u : \mathrm{id}_{\mathcal{C}} \implies RL \quad \text{and} \quad c : LR \implies \mathrm{id}_{\mathcal{D}},$$

called the *unit* and *counit* of the adjunction. Such transformations c and u give rise to an adjunction if and only if the *triangle identities*

$$(L \xRightarrow{Lu} LRL \xRightarrow{cL} L) = \mathrm{id}_L$$

$$(R \xRightarrow{uR} RLR \xRightarrow{Rc} R) = \mathrm{id}_R$$

hold.

Proof. MacLane ...

□

I.23. Remark

This latter description does not require $\mathcal{C}$ and $\mathcal{D}$ to be locally small so extends the notion of adjunction to arbitrary categories. In particular, one can check that an adjunction $L : \mathcal{C} \rightleftarrows \mathcal{D} : R$ induces $- \circ R : \mathrm{Fun}(\mathcal{C}, \mathcal{E}) \rightleftarrows \mathrm{Fun}(\mathcal{D}, \mathcal{E}) : - \circ L$ and $L \circ - : \mathrm{Fun}(\mathcal{E}, \mathcal{C}) \rightleftarrows \mathrm{Fun}(\mathcal{E}, \mathcal{D}) : R \circ -$ by extending the unit/counit in the evident way.

⁹The tautological transformation $\mathrm{Hom}_{\mathcal{C}} \Rightarrow \mathrm{Hom}_{\mathcal{D}} \circ (F^{\mathrm{op}} \times F)$ for any $F : \mathcal{C} \rightarrow \mathcal{D}$.

¹⁰See footnote ⁹.

I.24. Proposition

A left adjoint functor L is an embedding if and only if the unit of adjunction $u : \text{id} \Rightarrow RL$ is a natural isomorphism, and dually a right adjoint is an embedding if and only if the counit $c : LR \Rightarrow \text{id}$ is an isomorphism.

Proof. MacLane ...

□

I.25. Definition

A functor admitting a fully faithful left or right adjoint is called a right or left *Bousfield localisation*.

I.26. Remark

In general a *localisation* of a category $\mathcal{C}$ at a class of morphisms $S \subseteq \text{mor}(\mathcal{C})$ is a functor $L : \mathcal{C} \rightarrow \mathcal{D}$ such that for any category $\mathcal{E}$ the induced functor

$$- \circ L : \text{Fun}(\mathcal{D}, \mathcal{E}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{E})$$

is an embedding with essential image

$$\{F : \mathcal{C} \rightarrow \mathcal{E} \mid F(s) \text{ is an isomorphism for all } s \in S\}.$$

For any S and $\mathcal{C}$ there exists a contractible 2-groupoid of such localisations, but note that the Yoneda Lemma does not apply here! Furthermore, even if $\mathcal{C}$ is locally small the localisation, usually denoted $\mathcal{C}[S^{-1}]$ need not be so. However, any Bousfield localisation is a localisation (at which class of morphisms?), where the latter problem does not occur!

Here is another general example of adjoints:

I.27. Definition

Given functors $i : \mathcal{C} \rightarrow \mathcal{C}'$ and $F : \mathcal{C} \rightarrow \mathcal{D}$ a *left* or *right Kan extension* of F along i is a functor $\mathcal{C}' \rightarrow \mathcal{D}$ that is left or right adjoint to F under

$$- \circ i : \text{Fun}(\mathcal{C}', \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D}).$$

Kan extensions are denoted $\text{Lan}_i(F)/\text{Ran}_i(F)$ or $i_!F/i_*F$.

Decoding, a witness for G being a left Kan extension is a natural transformation $\sigma : F \Rightarrow G \circ i$ such that any natural transformation $\sigma' : F \Rightarrow G' \circ i$ factors uniquely through σ ¹¹:

¹¹Succinctly: $\text{Nat}(\text{Lan}_i F, G') \cong \text{Nat}(F, G' \circ i)$.

I.28. Examples

- i) For a group homomorphism $\varphi: H \rightarrow G$ the equivalence

$$\text{Fun}(\mathbb{B}G, \mathbf{R}\text{-Mod}) \simeq \mathbf{R}[G]\text{-Mod}$$

translates left/right Kan extension along

$$\varphi^*: \mathbf{R}[G]\text{-Mod} \rightarrow \mathbf{R}[H]\text{-Mod}$$

into induction and coinduction

$$M \mapsto R[G] \otimes_{R[H]} M \quad \text{and} \quad M \mapsto \text{Hom}_{R[H]}(R[G], M)$$

- ii) Given an additive functor $F: \mathbf{R}\text{-Mod} \rightarrow \mathbf{S}\text{-Mod}$ (or any other abelian categories with enough projectives), e.g. $- \otimes_R M$ or $\text{Hom}_R(M, -)$ the left derived functors $\mathbb{L}_i F$ are given by the right Kan extension

$$\begin{array}{ccc} \text{CC}_R^{\geq 0} & \xrightarrow{\quad} & \text{CC}_S^{\geq 0} \\ \pi_R \downarrow & \nearrow & \downarrow \pi_S \\ \mathcal{D}^{\geq 0}(R) & \xrightarrow{\text{Ran}_{\pi_R}(\pi_S \circ F)} & \mathcal{D}^{\geq 0}(S) \end{array}$$

known as the total left derived (or sometimes hyperhomology) functor $\mathbb{L}F$, composed with the functor $-[0]: \mathbf{R}\text{-Mod} \rightarrow \mathcal{D}(R)$, that regards a module as a single degree chain complex, and the homology functor $H_i: \mathcal{D}(R) \rightarrow \mathbf{R}\text{-Mod}$.

- iii) A homotopy invariant functor $F: \text{Top} \rightarrow \mathcal{C}$ is Kan extended from a homotopy invariant functor $\text{CW} \rightarrow \mathcal{C}$ if and only if F sends weak homotopy equivalences to isomorphisms.
- iv) For $i: \mathcal{C} \rightarrow \mathcal{C}'$, $i': \mathcal{C}' \rightarrow \mathcal{C}''$ and $F: \mathcal{C} \rightarrow \mathcal{D}$ we find straight from the definitions that

$$\text{Lan}_{i'}(\text{Lan}_i(F)) \simeq \text{Lan}_{i' \circ i}(F).$$

On the other hand we set

I.29. Definition

A functor $G: \mathcal{D} \rightarrow \mathcal{E}$ preserves the left/right Kan extension of $F: \mathcal{C} \rightarrow \mathcal{D}$ along $i: \mathcal{C} \rightarrow \mathcal{C}'$ if the diagram

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{F} \mathcal{D} & \xrightarrow{G} \mathcal{E} \\ \downarrow i & \searrow G \circ F & \nearrow G \circ \text{Lan}_i F \\ \mathcal{C}' & & \end{array} \quad \begin{array}{ccc} \mathcal{C} & \xrightarrow{F} \mathcal{D} & \xrightarrow{G} \mathcal{E} \\ \downarrow i & \searrow G \circ F & \nearrow G \circ \text{Ran}_i F \\ \mathcal{C}' & & \end{array}$$

exhibits $G \circ \text{Lan}_i F / G \circ \text{Ran}_i F$ as the left/right Kan extension of GF along i .

I.30. Proposition

Left adjoints preserve left Kan extensions and right adjoints preserve right Kan extensions.

Proof. Let $L: \mathcal{D} \rightarrow \mathcal{E}$ be left adjoint to $R: \mathcal{E} \rightarrow \mathcal{D}$. Then for $G: \mathcal{C}' \rightarrow \mathcal{E}$

$$\begin{aligned} \text{Nat}(L \text{Lan}_i F, G) &\cong^{12} \text{Nat}(\text{Lan}_i F, RG) \\ &\cong \text{Nat}(F, RGi) \\ &\cong \text{Nat}(LF, Gi) \\ &\cong \text{Nat}(\text{Lan}_i LF, G) \end{aligned}$$

and tracing through the definitions shows that for $G = L \text{Lan}_i F$ we find

$$\text{id}_{L \text{Lan}_i F} \mapsto^{13} u \circ \text{Lan}_i F \mapsto^{14} u \circ y \mapsto^{15} c \circ L(u \circ y) =^{16} Ly$$

where $y: F \Rightarrow (\text{Lan}_i F) \circ i$ is the original witness of the Kan extension. $\square$

I.31. Definition

A Kan extension is called *absolute* if it is preserved by all functors. It is called *pointwise* if it is preserved by representables: For a left/right Kan extension this means preservation by

$$(Y^*)^{\text{op}}: \mathcal{D} \rightarrow \text{Set}^{\text{op}} \quad \text{or} \quad Y_*: \mathcal{D} \rightarrow \text{Set},$$

respectively.

I.32. Proposition

If $i: \mathcal{C} \rightarrow \mathcal{C}'$ admits a right or left adjoint j then any functor $F: \mathcal{C} \rightarrow \mathcal{D}$ admits a left or right Kan extension, respectively. It is given by $F \circ j$ and thus absolute and in particular pointwise.

Note the order reversal, succinctly

$$\text{Lan}_L(F) = F \circ R \quad \text{and} \quad \text{Ran}_R(F) = F \circ L.$$

Proof. By Remark I.23 the functor

$$- \circ j: \text{Fun}(\mathcal{C}, \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}', \mathcal{D})$$

is left or right adjoint to $- \circ i$ if j is right or left adjoint to i . Thus $F \circ j$ satisfies the universal property of the Kan extension. $\square$

The adjunction witness determines isomorphisms

$$R \cong \text{Lan}_L(\text{id}_{\mathcal{C}}) \quad \text{and} \quad L \cong \text{Ran}_R(\text{id}_{\mathcal{D}})$$

¹²By I.23

¹³Definition of u .

¹⁴Definition of y .

¹⁵Definition of c .

¹⁶Triangle identity.

and these Kan extensions are absolute.

In fact, the converse holds as well: A functor R is right adjoint to L if and only if R is an absolute left Kan extension of $\text{id}_{\mathcal{C}}$ along L . The canonical map $\text{id}_{\mathcal{C}} \Rightarrow \text{Lan}_L(\text{id}_{\mathcal{C}}) \circ L$ is then exactly the counit of the adjunction.

As an important special case we also have:

I.33. Definition

Given a functor $F: \mathcal{J} \rightarrow \mathcal{D}$ the left and right Kan extensions of F along $r: \mathcal{J} \rightarrow *$ are called the *colimit* and *limit* of F , respectively, denoted $\text{colim } F$ ¹⁷ and $\text{lim } F$.

Let us unpack this again: Upon identifying $\text{Fun}(*, \mathcal{J}) \cong \mathcal{J}$, we find

$$\text{Hom}_{\mathcal{D}}(\text{colim } F, y) \cong \text{Nat}(F, \text{const}_{\mathcal{J}} y)$$

naturally in $y \in \mathcal{D}$, where $\text{const}_{\mathcal{J}} y$ is the constant functor with value y . The right hand side here consists of assignments $\tau: \text{obj}(\mathcal{J}) \rightarrow \text{mor}(\mathcal{D})$ with $\tau_x: F(x) \rightarrow y$ such that for all $f: x \rightarrow x'$ all triangles commute:

$$\begin{array}{ccc} F(x) & \xrightarrow{\tau_x} & y \\ f_* \downarrow & \nearrow \tau_{x'} & \\ F(x') & & \end{array}$$

The category $\mathcal{J}$ is often called the *shape* of the (co)limit.

I.34. Examples

- i) Colimits and limits of shape $\mathbb{D}S$ for $S \in \text{Set}$ are called *sums* (or *coproducts*) and *products*.
- ii) An element $x \in \mathcal{C}$ is called *initial* or *terminal* if it is a colimit or a limit, respectively, of the empty diagram $\mathbb{D}\emptyset \rightarrow \mathcal{C}$, i.e. if

$$\text{Hom}_{\mathcal{C}}(x, y) = \{*\} \quad \text{or} \quad \text{Hom}_{\mathcal{C}}(y, x) = \{*\}$$

for all $y \in \mathcal{C}$. Another formulation is that an object x is initial/terminal if and only if it is left/right adjoint to $r: \mathcal{C} \rightarrow *$, when regarded as a functor $* \rightarrow \mathcal{C}$.

- iii) The identification $\text{Fun}(\mathbb{B}G, \text{Set}) = G\text{-Set}$ translates limits and colimits into fixed points and orbits.
- iv) (Co)limits over the shapes $\mathbb{N}P$ for posets P can vary greatly: Colimits over $\mathbb{N}\{(0,0), (0,1), (1,0)\}$ with the evident order are called *pushouts* and limits over $\mathbb{N}\{(0,1), (1,0), (1,1)\}$ are called *pullbacks*.
- v) (Co)limits in functor categories.

From I.30 and I.32 we immediately obtain:

¹⁷Or informally $\text{colim}_{x \in \mathcal{C}} F(x)$

I.35. Corollary

- i) Left adjoint functors preserve colimits and right adjoints preserve limits.
- ii) If $\mathcal{J}$ admits an initial object $x \in \mathcal{J}$ then for every functor $F: \mathcal{J} \rightarrow \mathcal{C}$ we have $\lim F = F(x)$ and similarly $\operatorname{colim} F = F(y)$ if y is terminal in $\mathcal{J}$.

In particular, (co)limits as in ii) are absolute, which is fairly rare. We will see that the latter are precisely the colimits preserved by the Yoneda embedding.

We will need pointwise versions of I.23 and I.35 as well.

I.36. Observation

- i) Given $F: \mathcal{C} \rightarrow \mathcal{D}$ with partial left adjoint $L_F: \mathcal{D}_F \rightarrow \mathcal{C}$, and $G: \mathcal{E} \rightarrow \mathcal{D}$ with $\operatorname{im}(G) \subseteq \mathcal{D}_F$ then $L_F G \in \operatorname{Fun}(\mathcal{E}, \mathcal{C})$ is a left adjoint object to $G \in \operatorname{Fun}(\mathcal{E}, \mathcal{D})$ under

$$F \circ - : \operatorname{Fun}(\mathcal{E}, \mathcal{C}) \rightarrow \operatorname{Fun}(\mathcal{E}, \mathcal{D}).$$

- ii) Given $F: \mathcal{C} \rightarrow \mathcal{D}$ and $G: \mathcal{I} \rightarrow \mathcal{D}$ with $G(\mathcal{I}) \subseteq \mathcal{D}_F$ such that $\operatorname{colim}_{\mathcal{I}} G$ exists (in $\mathcal{D}$) with witness $y: G \Rightarrow \operatorname{const}_{\operatorname{colim} G}$. Then the following are equivalent:
 - a) $\operatorname{colim}_{\mathcal{I}} G \in \mathcal{D}_F$
 - b) $\operatorname{colim}_{\mathcal{I}} L_F G$ exists (in $\mathcal{C}$)

In this case the natural map

$$L_F G \xrightarrow{L_F y} L_F \operatorname{const}_{\operatorname{colim} G} = \operatorname{const}_{L_F \operatorname{colim} G}$$

exhibits $L_F \operatorname{colim}_{\mathcal{I}} G$ as a colimit of $L_F G$, i.e.

$$\operatorname{colim}_{\mathcal{I}} L_F G = L_F(\operatorname{colim}_{\mathcal{I}} G).$$

Similar statements hold for $\mathcal{D}^F, R^F$, right adjoints and limits.

We use this observation to establish another basic property of colimits:

I.37. Proposition

Given $G: \mathcal{I} \times \mathcal{J} \rightarrow \mathcal{C}$ denote by $G_0: \mathcal{I} \rightarrow \operatorname{Fun}(\mathcal{J}, \mathcal{C})$ and $G_1: \mathcal{J} \rightarrow \operatorname{Fun}(\mathcal{I}, \mathcal{C})$ the curried functors. Then

- i) If $G_1(j): \mathcal{I} \rightarrow \mathcal{C}$ admits a colimit for all $j \in \mathcal{J}$, then G_0 admits a colimit and the canonical map

$$\operatorname{colim}_{\mathcal{I}} G_1(j) \rightarrow (\operatorname{colim}_{\mathcal{I}} G_0)(j)$$

is an isomorphism for all $j \in \mathcal{J}$; similarly for limits.

- ii) If $G_0(i): \mathcal{J} \rightarrow \mathcal{C}$ admits a colimit for each $i \in \mathcal{I}$ and $G_1(j): \mathcal{I} \rightarrow \mathcal{C}$ admits a colimit for each $j \in \mathcal{J}$, then

$$\operatorname{colim}_{\mathcal{I}}(\operatorname{colim}_{\mathcal{J}} G_1) \text{ exists} \quad \text{iff} \quad \operatorname{colim}_{\mathcal{J}}(\operatorname{colim}_{\mathcal{I}} G_0) \text{ exists},$$

in which case they agree; similarly for limits.

One loosely says that "(co)limits in functor categories are computed pointwise", although this only applies to diagrams whose colimit *can* be computed pointwise: It can happen that $\operatorname{colim}_{\mathcal{I}} G_0$ exists "accidentally" without the colimits that could define its values existing. Similarly for part ii) one often says "colimits commute". The value is also given by $\operatorname{colim}_{\mathcal{I} \times \mathcal{J}} G$ but again this latter object may exist without either $\operatorname{colim}_{\mathcal{I}} G_0$ or $\operatorname{colim}_{\mathcal{J}} G_1$ existing.

Proof. Observe that by assumption

$$G_1: \mathcal{J} \rightarrow \operatorname{Fun}(\mathcal{I}, \mathcal{C})$$

takes values in $\operatorname{Fun}(\mathcal{I}, \mathcal{C})_{r_{\mathcal{I}}^*}$ with $r_{\mathcal{I}}^*: \mathcal{C} \rightarrow \operatorname{Fun}(\mathcal{I}, \mathcal{C})$ the "constant diagram" functor, as this latter category consists precisely of those functors admitting colimits over $\mathcal{I}$ (by definition of colimits!) Thus there is the partial adjoint

$$\operatorname{colim}_{\mathcal{I}}: \operatorname{Fun}(\mathcal{I}, \mathcal{C})_{r_{\mathcal{I}}^*} \rightarrow \mathcal{C}$$

and part i) of I.36 implies that $\operatorname{colim}_{\mathcal{I}} \circ G_1: \mathcal{J} \rightarrow \mathcal{C}$ is a left adjoint object to G_1 under

$$r_{\mathcal{I}}^* \circ - : \operatorname{Fun}(\mathcal{J}, \mathcal{C}) \rightarrow \operatorname{Fun}(\mathcal{J}, \operatorname{Fun}(\mathcal{I}, \mathcal{C})).$$

But using the equivalence

$$\operatorname{Fun}(\mathcal{J}, \operatorname{Fun}(\mathcal{I}, \mathcal{C})) \simeq \operatorname{Fun}(\mathcal{I} \times \mathcal{J}, \mathcal{C}) \simeq \operatorname{Fun}(\mathcal{I}, \operatorname{Fun}(\mathcal{J}, \mathcal{C}))$$

$r_{\mathcal{I}}^* \circ -$ corresponds to $- \circ r_{\mathcal{I}}$, and G_1 to G_0 , so $\operatorname{colim}_{\mathcal{I}} \circ G_1$ is a colimit of G_0 and clearly satisfies the desired formula. This gives i). For ii) we use part ii) of I.36 with $Fr_{\mathcal{I}}^*: \mathcal{C} \rightarrow \operatorname{Fun}(\mathcal{I}, \mathcal{C})$ and $L_F = \operatorname{colim}_{\mathcal{I}}$:

If $G_1: \mathcal{J} \rightarrow \operatorname{Fun}(\mathcal{I}, \mathcal{C})$ takes values in $\operatorname{Fun}(\mathcal{I}, \mathcal{C})_{r_{\mathcal{I}}^*}$, $\operatorname{colim}_{\mathcal{J}} G_1$ exists (which follows from the assumption by part i)) and $\operatorname{colim}_{\mathcal{I}}(\operatorname{colim}_{\mathcal{J}} G_1)$ exists then also $\operatorname{colim}_{\mathcal{I}} \circ G_1$ exists and

$$\operatorname{colim}_{\mathcal{I}}(\operatorname{colim}_{\mathcal{J}} G_1) = {}^{18} \operatorname{colim}_{\mathcal{J}}(\operatorname{colim}_{\mathcal{I}} \circ G_1) = {}^{19} \operatorname{colim}_{\mathcal{J}}(\operatorname{colim}_{\mathcal{I}} G_0)$$

□

I.38. Definition

A category $\mathcal{C}$ is called *complete* or *cocomplete* if it admits all limits or colimits of small shape, respectively.

All the usual examples Grp, Set, Top, R-Mod, ... are both!

¹⁸By part ii) of I.35

¹⁹By i)

I.39. Corollary

If $\mathcal{D}$ is (co)complete then so is $\text{Fun}(\mathcal{C}, \mathcal{D})$ for every category $\mathcal{C}$, in particular $\mathbf{P}(\mathcal{C})$ is both complete and cocomplete for every $\mathcal{C}$. Furthermore, for any $F: \mathcal{C} \rightarrow \mathcal{C}'$ the functor

$$F^*: \text{Fun}(\mathcal{C}', \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$$

preserves all limits and colimits.²⁰

In fact, we will see below that under a further smallness assumption F^* admits both adjoints $F_!$ and F_* (a.k.a. $\text{Lan}_F(-)$ and $\text{Ran}_F(-)$) from which the last statement also follows in that case.

Now we have:

I.40. Observation

The functors $Y^x: \mathcal{C}^{\text{op}} \rightarrow \text{Set}$ and $Y_x: \mathcal{C} \rightarrow \text{Set}$ preserve small limits for every (left locally small) $x \in \mathcal{C}$.²¹ For locally small $\mathcal{C}$ therefore the Yoneda embeddings

$$Y^{\mathcal{C}}: \mathcal{C} \rightarrow \mathbf{P}(\mathcal{C}) \quad \text{and} \quad Y_{\mathcal{C}}: \mathcal{C}^{\text{op}} \rightarrow \mathbf{P}(\mathcal{C}^{\text{op}})$$

preserve all limits.

Proof. We note that

$$\text{Hom}_{\mathcal{C}}(x, \lim_{\mathcal{I}} F) = \text{Nat}(\text{const}_{\mathcal{I}} x, F)$$

where the latter by inspection is $\lim_{\mathcal{I}} \text{Hom}_{\mathcal{C}}(x, F(-))$, with both consisting of assignments

$$\tau: \text{obj}(\mathcal{I}) \rightarrow \text{mor}(\mathcal{C})$$

with $\tau_i: x \rightarrow F(i)$, such that for all $f: i \rightarrow j \in \mathcal{I}$ the diagram

$$\begin{array}{ccc} x & \xrightarrow{\tau_i} & F(i) \\ \tau_j \downarrow & \swarrow & \downarrow F(f) \\ & & F(j) \end{array}$$

commutes.

The first statement (for Y_x) follows, since

$$Y_x(\lim F) = \text{Hom}_{\mathcal{C}}(x, \lim F) = \lim_{\mathcal{C}} \text{Hom}(x, F(-)) = \lim_{\mathcal{C}} Y^x \circ F$$

The second one is similar:

$$Y^{\mathcal{C}}(\lim F) = \text{Hom}_{\mathcal{C}}(-, \lim F) = \lim_{\mathcal{C}} \text{Hom}_{\mathcal{C}}(-, F) = \lim Y^{\mathcal{C}} \circ F$$

□

Colimits are very much *not* preserved by the Yoneda embedding! Before we analyse the situation, let's justify the smallness restriction in the definition of completeness.

²⁰Because they are pointwise!

²¹I.e. $\text{Hom}_{\mathcal{C}}(\text{colim}_{\mathcal{I}} F, y) = \lim_{\mathcal{I}^{\text{op}}} \text{Hom}(F(-), y) = \text{Hom}_{\mathcal{C}^{\text{op}}} (y, \lim_{\mathcal{I}^{\text{op}}} F^{\text{op}})$

²²Uses the fact that limits in $\mathbf{PC}$ are pointwise!

I.41. Proposition (Freyd)

If a category $\mathcal{C}$ admits coproducts for all diagrams of shape $\mathbb{D}(\text{mor}(\mathcal{C}))$, then $\mathcal{C}$ is equivalent to $N(P)$ for some poset P . In particular, small cocomplete categories are posets.

Proof. Let $x, y \in \mathcal{C}$ be two objects. Then form the sum $\sum_{\mathbb{D}(\text{mor}(\mathcal{C}))} x$, which exists by assumption. Then

$$\text{Hom}_{\mathcal{C}}\left(\sum_{\text{mor}(\mathcal{C})} x, y\right) = \prod_{\text{mor}(\mathcal{C})} \text{Hom}_{\mathcal{C}}(x, y)$$

where the former has cardinality at most that of $\text{mor}(\mathcal{C})$, whereas the latter has cardinality at least that of the power set of $\text{mor}(\mathcal{C})$, unless $|\text{Hom}_{\mathcal{C}}(x, y)| \leq 1$. □

We are now ready to state the main theorem of this section:

I.42. Theorem

If $\mathcal{D}$ is cocomplete, and $\mathcal{C}$ small, then restriction along the Yoneda embedding induces an equivalence

$$(Y^{\mathcal{C}})^* : \text{Fun}^{\text{colim}}(\text{P}(\mathcal{C}), \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$$

where the superscript denotes the full subcategory on those functors that preserve all small colimits. The inverse functor is given by left Kan extension along $Y^{\mathcal{C}}$.

In other words, $\text{P}(\mathcal{C})$ is left 2-adjoint²³ to $\mathcal{C}$ under the inclusion of cocomplete categories with colimit preserving functors into all categories.

In particular,

$$\text{Fun}^{\text{colim}}(\text{Set}, \mathcal{C}) \simeq \mathcal{C}$$

means that Set is freely generated under colimits by any one-point set. Generally, $\text{P}(\mathcal{C})$ is the free cocompletion of $\mathcal{C}$.

The proof needs a few preparations:

I.43. Definition

Given functors $F: \mathcal{C} \rightarrow \mathcal{D}$ and $G: \mathcal{C}' \rightarrow \mathcal{D}$ define the *slice category* F/G by

$$\text{obj}(F/G) = \{(x, y, f) \in \text{obj}(\mathcal{C}) \times \text{obj}(\mathcal{C}') \times \text{mor}(\mathcal{D}) \mid f: F(x) \rightarrow G(y)\}$$

$$\text{Hom}_{F/G}((x, y, f), (x', y', f')) = \left\{ (g, g') \in \text{mor}(\mathcal{C}) \times \text{mor}(\mathcal{C}') \mid \begin{array}{c} \begin{array}{l} \bullet g: x \rightarrow y \\ \bullet g': x' \rightarrow y' \end{array} \\ F(x) \xrightarrow{f} G(y) \\ \begin{array}{ccc} F(g) \downarrow & & \downarrow G(g') \\ F(x') & \xrightarrow{f'} & G(y') \end{array} \end{array} \right\}$$

If $F = \text{id}_{\mathcal{C}}$ one writes $\mathcal{C}/G$, and if $F: * \rightarrow \mathcal{C}$ one writes $F(*)/G$. As special cases we find

²³I.e. in the 2-categorical sense, because $(Y^{\mathcal{C}})^*$ is an equivalence, but not an iso!

- $\mathcal{D}/\mathcal{D} = \text{Fun}(\mathbb{N}[1], \mathcal{C})$, the *arrow category* of $\mathcal{D}$.
- $d/\mathcal{D}$ the *under category* of $d \in \mathcal{D}$:

$$\text{obj}(d/\mathcal{D}) = \{(x, f) \mid f: d \rightarrow \mathcal{D}\}$$

$$\text{Hom}_{d/\mathcal{D}}((x, f), (y, g)) = \left\{ \varphi: x \rightarrow y \mid \begin{array}{c} d \xrightarrow{f} x \\ g \downarrow \swarrow \varphi \\ y \end{array} \right\}$$

and similarly the *overcategory* $\mathcal{D}/d$.

There are always forgetful functors

$$\begin{array}{ccc} s: F/G \rightarrow \mathcal{C} & & t: F/G \rightarrow \mathcal{C}' \\ (x, y, f) \mapsto x & \text{and} & (x, y, f) \mapsto y \end{array}$$

and a tautological natural transformation

$$\begin{array}{c} \tau: Fs \Rightarrow Gt \\ \tau(x, y, f) = f \end{array}$$

Armed with this definition, we turn to the existence of Kan extensions:

I.44. Proposition

Given $i: \mathcal{C} \rightarrow \mathcal{C}'$ and $F: \mathcal{C} \rightarrow \mathcal{D}$ then the Kan extension $\text{Lan}_i F$ is pointwise if and only if the canonical map

$$\text{colim}(i/c' \xrightarrow{s} \mathcal{C} \xrightarrow{F} \mathcal{D}) \rightarrow (\text{Lan}_i F)(c')$$

is an isomorphism for all $c' \in \mathcal{C}'$. Dually, $\text{Ran}_i F$ is pointwise if and only if the natural map

$$(\text{Ran}_i F)(c') \rightarrow \lim(c'/i \xrightarrow{t} \mathcal{C} \xrightarrow{F} \mathcal{D})$$

is an isomorphism.

If conversely the displayed (co)limits exist for all $c' \in \mathcal{C}'$, then they combine into Kan extensions of F along i (which are then automatically pointwise).

I.45. Corollary

If $\mathcal{C}$ is small, $\mathcal{C}'$ locally small²⁵ and $\mathcal{D}$ cocomplete or complete, respectively, then all functors $\mathcal{C} \rightarrow \mathcal{D}$ admit left or right Kan extensions along i , all of which are pointwise. In particular,

$$i^*: \text{Fun}(\mathcal{C}', \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$$

admits both adjoints in this case (namely Lan_i and Ran_i).

²⁴ $\text{obj}(i/c') = \{(c, f) \mid f: i(c) \rightarrow c'\}$

²⁵Makes i/c' and c'/i small for all $c' \in \mathcal{C}'$.

I.46. Proof of I.44

Assume that $\text{Ran}_i F$ is pointwise. recall that this means that

$$Y_d \circ \text{Ran}_i F = \text{Ran}_i(Y_d \circ F)$$

i.e. the following diagram commutes:

$$\begin{array}{ccccc} \mathcal{C} & \xrightarrow{\quad} & \mathcal{D} & \xrightarrow{\text{Hom}(d, -)} & \text{Set} \\ & \searrow \text{Ran}_i F & & \nearrow & \\ i \downarrow & & & & \\ \mathcal{C}' & & & \nearrow \text{Ran}_i(\text{Hom}(d, F(-))) & \end{array}$$

Then we compute

$$\begin{aligned} \text{Hom}_{\mathcal{D}}(d, (\text{Ran}_i F)(i')) &= (Y_d \circ \text{Ran}_i F)(c') \\ &= \text{Nat}(Y_{c'}, Y_d \circ (\text{Ran}_i F)) \\ &= \text{Nat}(Y_{c'}, \text{Ran}_i(Y_d \circ F)) \\ &= \text{Nat}(Y_{c' \circ i}, Y_d \circ F) \\ &= \text{Nat}(\text{Hom}_{\mathcal{C}'}(c', i(-)), \text{Hom}_{\mathcal{D}}(d, F(-))). \end{aligned}$$

Now

$$\text{Nat}(\text{Hom}_{\mathcal{C}'}(c', i(-)), \text{Hom}_{\mathcal{D}}(d, F(-)))$$

consists of maps

$$\tau : \text{obj}(\mathcal{C}) \rightarrow \text{mor}(\text{Set})$$

$$\tau_c : \text{Hom}_{\mathcal{C}'}(c', i(c)) \rightarrow \text{Hom}_{\mathcal{D}}(d, Fc)$$

that are natural in c . Such a τ is exactly the same map as a map

$$\text{obj}(c'/i) \rightarrow \text{mor}(\text{Set})$$

that forms a natural transformation

$$\text{const}_{c'/i} d \Rightarrow F \circ t$$

Thus

$$\text{Hom}_{\mathcal{D}}(d, (\text{Ran}_i F)(c')) = \text{Nat}(\text{const}_{c'/i} d, F \circ t)$$

which exhibits $(\text{Ran}_i F)(c')$ as a limit of $F \circ t$, since $d \in \mathcal{D}$ was arbitrary.

Conversely, if all the functors $c'/i \xrightarrow{t} \mathcal{C} \xrightarrow{F} \mathcal{D}$ admit limits for all $c' \in \mathcal{C}'$, then they fit together into a functor $R: \mathcal{C}' \rightarrow \mathcal{D}$, with morphism component on $g: c' \rightarrow c''$ induced by

$$\begin{aligned} g^* : c''/i &\rightarrow c'/i \\ (x, f) &\mapsto (x, f \circ g) \end{aligned}$$

(note $F \circ t_{c'} \circ g^* = F \circ t_{c''}$) and the canonical map

$$\lim_{\mathcal{D}} A \rightarrow \lim_{\mathcal{C}} (A \circ B)$$

for functors $B: \mathcal{C} \rightarrow \mathcal{D}$, and $A: \mathcal{D} \rightarrow \mathcal{E}$.

R acquires a natural transformation

$$\lambda: R \circ i \Rightarrow F$$

by projecting $\lim^{i(c)/i} \xrightarrow{t} \mathcal{C} \xrightarrow{F} \mathcal{D}$ to the component of $(c, \text{id}_{i(c)}) \in i(c)/i$. Given another functor and transformation $R' \circ i \xRightarrow{\tau} F$, then we obtain elements in

$$\text{Hom}_{\mathcal{D}}(R'(c'), R(c')) = \text{Nat}(\text{const}_{c'/i} R'(c'), F \circ t)$$

by

$$(x, f: c' \rightarrow i(x)) \mapsto (R'(c') \xrightarrow{f_*} R'(i(x)) \xrightarrow{\tau_x} F(x)).$$

These fit into a natural transformation $\sigma: R' \Rightarrow R$ that makes

$$\begin{array}{ccc} R'i & \xrightarrow{\sigma_i} & Ri \\ \tau \downarrow & \swarrow \lambda & \\ F & & \end{array}$$

commutative (by unwrapping the definitions). We leave it to the reader to check that σ is unique with this property, which shows R to be a right Kan extension of F along i .

That it is pointwise follows immediately from I.40. $\square$

Corollary I.45 yields the functor

$$\text{Lan}_{Y^c}: \text{Fun}(\mathcal{C}, \mathcal{D}) \rightarrow \text{Fun}(\text{P}(\mathcal{C}), \mathcal{D})^{26}$$

which we claim is inverse to the restriction in I.42. We first check that its essential image is indeed contained in $\text{Fun}^{\text{colim}}(\text{P}(\mathcal{C}), \mathcal{D})$: Since the Kan extension is pointwise and colimits in $\text{P}(\mathcal{C})$ are computed pointwise, this is immediate from part ii) of I.37 and I.44.

To see that Lan_{Y^c} really is an inverse we have:

I.47. Proposition

If $i: \mathcal{C} \rightarrow \mathcal{C}'$ is an embedding and $F: \mathcal{C} \rightarrow \mathcal{D}$ a functor with pointwise left/right Kan extension, then the structure maps

$$F \Rightarrow (\text{Lan}_i F) \circ i \quad \text{and} \quad (\text{Ran}_i F) \circ i \Rightarrow F$$

are natural equivalences.

In this case $\text{Lan}_i F$ and $\text{Ran}_i F$ really are extensions of F .

Proof. We saw (in I.46) that the structural map

$$\lambda_c: (\text{Ran}_i F)(i(c)) \rightarrow F(c)$$

is given by selecting the component of $(c, \text{id}_{i(c)}) \in i(c)/i$ to give a map

$$(\text{Ran}_i F)(i(c)) = \lim^{i(c)/i} \xrightarrow{t} \mathcal{C} \xrightarrow{F} \mathcal{D} \rightarrow F(c).$$

²⁶ $\mathcal{D}$ is cocomplete.

But if i is an embedding then $(c, \text{id}_{i(c)})$ is initial in $i(c)/i$: A map

$$(c, \text{id}_{i(c)}) \rightarrow (x, f: i(c) \rightarrow i(x))$$

consists of a map $g: c \rightarrow x$ such that

$$\begin{array}{ccc} i(c) & \xrightarrow{f} & i(x) \\ \text{id}_{i(c)} \downarrow & \nearrow i(g) & \\ i(c) & & \end{array}$$

commutes. Thus $f = i(g)$ which determines g uniquely. Thus by part ii) of I.35 λ_c is an isomorphism and by part ii) of I.14 we are done. $\square$

In particular, this applies to Lan_{Y^C} and shows that the composite

$$\text{Fun}(\mathcal{C}, \mathcal{D}) \xrightarrow{\text{Lan}_{Y^C}} \text{Fun}(\text{P}(\mathcal{C}), \mathcal{D}) \xrightarrow{(Y^C)^*} \text{Fun}(\mathcal{C}, \mathcal{D})$$

is equivalent to the identity. To see the other direction we have:

I.48. Proposition

For every $F \in \text{P}(\mathcal{C})$ the tautological map

$$\text{colim}(Y^C/F \xrightarrow{s} \mathcal{C} \xrightarrow{Y^C} \text{P}(\mathcal{C})) \Rightarrow F$$

induced by the tautological transformation

$$\begin{array}{ccc} Y^C/F & \xrightarrow{s} & \mathcal{C} \xrightarrow{Y^C} \text{P}(\mathcal{C}) \\ & \Downarrow & \\ Y^C/F & \xrightarrow{t} & * \xrightarrow{F} \text{P}(\mathcal{C}) \end{array}$$

is an isomorphism.

Said more abstractly: The left Kan extension of Y^C along itself is the identity of $\text{P}(\mathcal{C})$. The category Y^C/F is often called the *category of objects* of F : By definition it consists of pairs (x, f) with $f: Y^x \Rightarrow F$. By Yoneda's Lemma this is equivalent to a pair (x, f) with $f \in F(X)$. Similarly, there is an identification

$$\text{Hom}((x, f), (y, g)) \cong \left\{ \varphi: x \rightarrow y \mid \varphi^* \widehat{g} = \widehat{f} \right\}.$$

The category Y^C/F is the simplest example of a Grothendieck construction, which Lurie's unstraightening construction generalises to ∞ -categories.

For us the important observation is that *every* $F \in \text{P}(\mathcal{C})$ is a colimit of representables. In particular, it illustrates the failure of Y^C to commute with colimits quite drastically.

Proof. We compute for $G: \mathcal{C}^{\text{op}} \rightarrow \text{Set}$

$$\begin{aligned} \text{Nat}(\text{colim}(Y^C/F \xrightarrow{s} \mathcal{C} \xrightarrow{Y^C} \text{P}(\mathcal{C})), G) &= \\ &= \lim_{(x, f) \in (Y^C/F)^{\text{op}}} \text{Nat}(Y^x, G) \\ &= \lim_{(x, f) \in (Y^C/F)^{\text{op}}} G(x) \\ &= \lim((Y^C/F)^{\text{op}} \xrightarrow{s} \mathcal{C}^{\text{op}} \xrightarrow{G} \text{Set}) \end{aligned}$$

But this limit consists of assignments

$$\tau : \text{obj}(Y^{\mathcal{C}}/F) \rightarrow \text{obj}(\text{Set})$$

$$\tau(x, f) \in G(x)$$

such that for $\varphi: y \rightarrow x$ we have

$$\tau(y, \varphi^* f) = \varphi^* \tau(x, f).$$

This is the same data as that of a natural transformation $\tau: F \Rightarrow G$. Thus

$$\text{Nat}(\text{colim}(Y^{\mathcal{C}}/F \xrightarrow{s} \mathcal{C} \xrightarrow{Y^{\mathcal{C}}} P(\mathcal{C})), G) = \text{Nat}(F, G)$$

and the claim follows from Yoneda's Lemma after unwinding. $\square$

To prove I.42 we finally need the following lemma:

I.49. Lemma

Let $L: \mathcal{C} \rightleftarrows \mathcal{D}: R$ be an adjoint pair of functors. If

- i) L is an embedding, and
- ii) R detects isomorphisms, that is if Rf is an isomorphism in $\mathcal{D}$ then f is one in $\mathcal{C}$,

then L and R are inverse equivalences of categories, witnessed by the unit and counit of the adjunction.

Proof. As L is an embedding the unit $u: \text{id}_{\mathcal{C}} \Rightarrow RL$ is a natural isomorphism by I.24. To check that the counit

$$c: LP \Rightarrow \text{id}_{\mathcal{D}}$$

is an isomorphism as well, we may apply R (as natural isomorphisms are detected pointwise, and R detects these). But the triangle identity

$$(R \xrightarrow{uR} RLR \xrightarrow{Rc} R) = \text{id}_R$$

tells us that Rc is left-inverse to an isomorphism so one itself. $\square$

Finally we have:

I.50. Proof of I.42

We checked that

$$\text{Lan}_{Y^{\mathcal{C}}} : \text{Fun}(\mathcal{C}, \mathcal{D}) \rightarrow \text{Fun}(P(\mathcal{C}), \mathcal{D})$$

is an embedding as an application of I.46, and so the same holds for its restriction

$$\text{Lan}_{Y^{\mathcal{C}}} : \text{Fun}(\mathcal{C}, \mathcal{D}) \rightarrow \text{Fun}^{\text{colim}}(P(\mathcal{C}), \mathcal{D})$$

as the right hand side was defined as a full subcategory. We are left to check that the restriction functor

$$Y_{\mathcal{C}}^* : \text{Fun}^{\text{colim}}(P(\mathcal{C}), \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$$

detects isomorphisms. This follows from I.47 as for a natural transformation $y: \mathcal{I} \Rightarrow \mathcal{J}$ of colimit-preserving functors $\mathbf{P}(\mathcal{C}) \rightarrow \mathcal{D}$, we have

$$\begin{array}{ccccc} \mathcal{I}(F) = \mathcal{I}\left(\operatorname{colim}_{(x,f) \in Y^{\mathcal{C}}/F} Y^x\right) & = & \operatorname{colim}_{(x,f) \in Y^{\mathcal{C}}/F} \mathcal{I}(Y^x) & & \\ Y_F \downarrow & & \downarrow & & \downarrow \operatorname{colim} y(Y^x) \\ \mathcal{J}(F) & = & \mathcal{J}(\operatorname{colim} Y^x) & = & \operatorname{colim} \mathcal{J}(Y^x) \end{array}$$

so if y is an isomorphism on representable it is an isomorphism. □

Chapter II.

Simplicial Sets